FROM ZERO TO PRODUCTION

Apache Kafka

The Complete Beginner-to-Advanced Guide
for Data Engineers

Python | Spark | Docker | Interview Ready

16 Chapters · Diagrams · Code Labs · Hands-on Projects · Cheat Sheet

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CHAPTER 1

Introduction to Apache Kafka

Understanding what Kafka is, why it exists, and where it fits in the real world.

What is Apache Kafka?

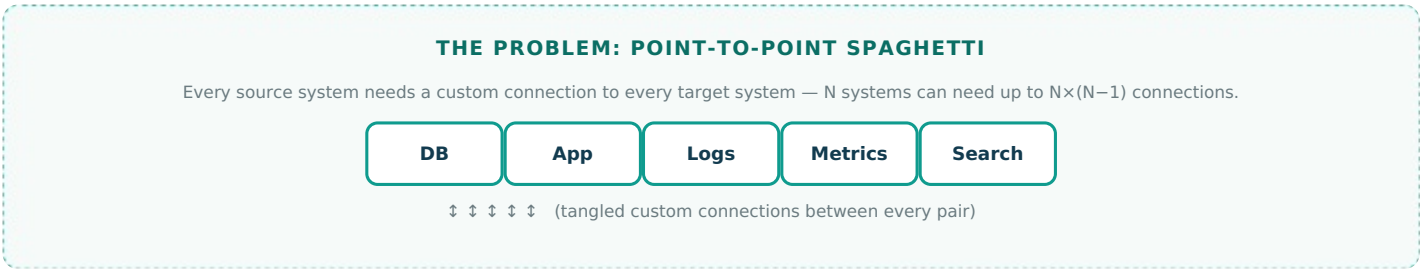
Apache Kafka is a **distributed event streaming platform**. In plain English: it is a system that lets applications send, store, and process huge streams of data (called "events") in real time, reliably, and at massive scale.

Think of Kafka as a super-powered **postal service for data**. One application "mails" a piece of information (an event), Kafka stores it safely in order, and any number of other applications can "read the mail" — instantly or hours later — without the sender ever needing to know who's reading it.

> **NOTE**
> An "event" is simply a fact that happened: a click, a payment, a sensor reading, a login, an order being placed. Kafka's job is to move and store these facts reliably.

Why Was Kafka Created?

Kafka was built at **LinkedIn in 2010–2011** by Jay Kreps, Neha Narkhede, and Jun Rao. LinkedIn had hundreds of systems (user activity tracking, monitoring, databases, search) that all needed to exchange data with each other. Connecting every system directly to every other system created a tangled, unmanageable mess.



LinkedIn needed one central system that every application could publish data to and every application could subscribe from — a single, reliable, scalable "nervous system" for company-wide data. That system became Kafka, later open-sourced under the Apache Software Foundation.

Problems Kafka Solves

Problem	How Kafka Solves It
Too many direct point-to-point integrations	One central hub — producers and consumers only need to know about Kafka, not each other
Systems crash or go offline	Kafka stores data durably on disk, so no events are lost while a consumer is down
Data volume grows faster than one server can handle	Kafka scales horizontally across many machines (a cluster)
Different systems need the same data at different speeds	Multiple independent consumers can read the same data at their own pace
Need for real-time decisions (fraud detection, alerts)	Kafka delivers events with millisecond-level latency

Real-World Use Cases

Industry	Use Case
🛒 E-commerce	Order processing pipelines, inventory updates, real-time recommendations
🏦 Banking & Fintech	Fraud detection, transaction streaming, real-time balance updates
🚗 Ride-hailing (Uber, Careem)	Live driver location tracking, surge pricing calculations
📺 Streaming (Netflix)	Viewing activity tracking, recommendation engines, monitoring
📈 Finance / Trading	Real-time stock price feeds and trade matching
🌐 IoT	Millions of sensors streaming temperature, location, or usage data
🔍 Log Aggregation	Centralizing logs from thousands of microservices for monitoring

> 💡 **REMEMBER**
>
> Kafka does not "process" data by itself by default — it **moves and stores** streams of events reliably. Processing is done by producers, consumers, or stream-processing tools like Kafka Streams or Spark.

🧳 Chapter 1 Summary

- Kafka is a distributed event streaming platform — a real-time, durable pipeline for data.
- It was created at LinkedIn to replace messy point-to-point integrations with one central hub.
- Kafka solves problems of scale, reliability, and real-time delivery of data between systems.
- It powers real-time systems at companies like LinkedIn, Uber, Netflix, and thousands of banks and e-commerce platforms.

🧠 Mini Quiz

Q1. What company originally created Kafka, and why?

A: LinkedIn, to replace tangled point-to-point integrations with one central data hub.

Q2. Does Kafka process data by default, or just move and store it?

A: It moves and stores data reliably; processing is done by separate producer/consumer/stream apps.

Q3. Name two real-world industries that rely heavily on Kafka.

A: Any two of: e-commerce, banking/fintech, ride-hailing, streaming media, IoT, log aggregation.

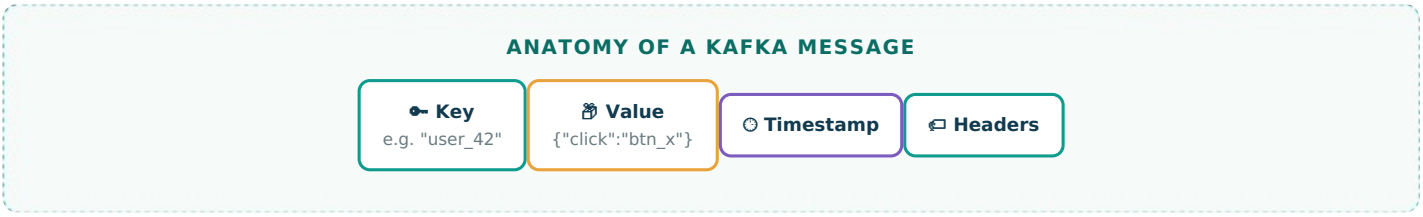
CHAPTER 2

🧩 Core Concepts

The vocabulary you must know before anything else in Kafka makes sense.

Event & Message

An **event** is a fact that something happened — "user_42 clicked button X at 10:03:12". A **message** is the technical package Kafka uses to carry that event: it has an optional **key**, a **value** (the actual data, often JSON or Avro), a **timestamp**, and optional **headers**. In everyday conversation, "event" and "message" are used almost interchangeably.



Topic

A **topic** is a named category or feed where messages are stored — similar to a table in a database, or a folder for a specific type of data. Examples: `orders`, `page_views`, `sensor_temperature`.

Partition

Every topic is split into one or more **partitions**. A partition is an ordered, append-only log of messages. Splitting a topic into partitions is what allows Kafka to scale — different partitions can live on different machines and be read in parallel.

> 💡 **REMEMBER**
> Kafka guarantees message **order only within a single partition** — never across the whole topic.

Offset

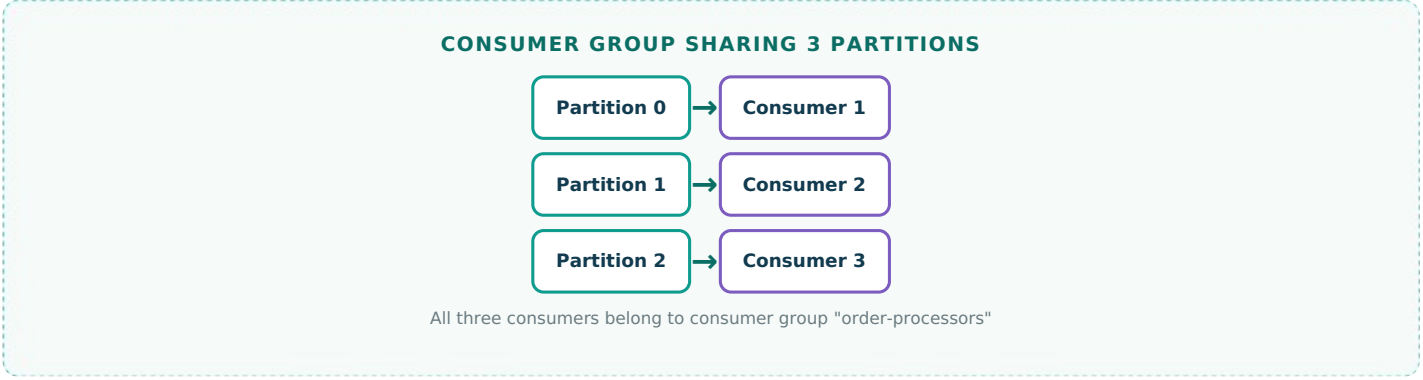
Each message inside a partition gets a unique, sequential ID number called an **offset** (0, 1, 2, 3...). Offsets let consumers keep track of exactly which messages they have already read.

Producer & Consumer

Role	What It Does	Real-World Analogy
👨‍💻 Producer	An application that writes/publishes messages to a topic	A person dropping a letter into a mailbox
🕵️ Consumer	An application that reads/subscribes to messages from a topic	A person checking their mailbox for new letters

Consumer Group

A **consumer group** is a set of consumers that work together to read a topic, sharing the partitions between them so each message is processed only once *per group*. This is how Kafka achieves parallel processing.



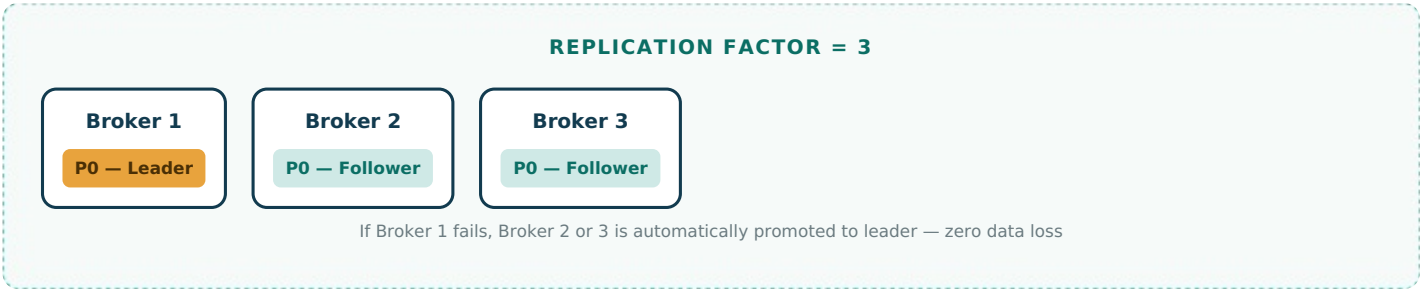
> ⚠ **COMMON MISTAKE**
>
> Adding more consumers to a group than there are partitions doesn't help — the extra consumers will sit **idle**. Max useful consumers per group = number of partitions.

Broker & Cluster

A **broker** is a single Kafka server — it stores data and serves client requests. A **cluster** is a group of brokers working together, coordinating to store and replicate data reliably.

Replication, Leader, and Follower Partitions

Replication means Kafka keeps multiple copies of each partition on different brokers, so if one broker fails, no data is lost. Each replicated partition has one **leader** (handles all reads/writes) and one or more **followers** (passively copy the leader's data, ready to take over if the leader fails).



Core Concepts — Quick Glossary

Term	One-Line Definition
Event / Message	A single piece of data representing something that happened
Topic	A named stream/category of messages
Partition	An ordered sub-log of a topic, enabling parallelism
Offset	The position/ID of a message within a partition
Producer	Application that writes messages to a topic
Consumer	Application that reads messages from a topic
Consumer Group	A team of consumers sharing the work of reading a topic
Broker	A single Kafka server
Cluster	A group of brokers working together
Replication	Keeping copies of data across brokers for fault tolerance
Leader / Follower	The active copy vs. the backup copies of a partition

🛍 Chapter 2 Summary

- Topics hold messages; topics are split into partitions for scale and parallelism.
- Order is guaranteed only within a partition, tracked using offsets.
- Producers write, consumers read; consumer groups split the reading work.
- Brokers form a cluster; replication with leader/follower partitions provides fault tolerance.

🧠 Mini Quiz

Q1. Is message order guaranteed across an entire topic?

A: No — only within a single partition.

Q2. What happens if you add more consumers than partitions to a group?

A: The extra consumers stay idle and receive no messages.

Q3. What is the difference between a leader and a follower partition?

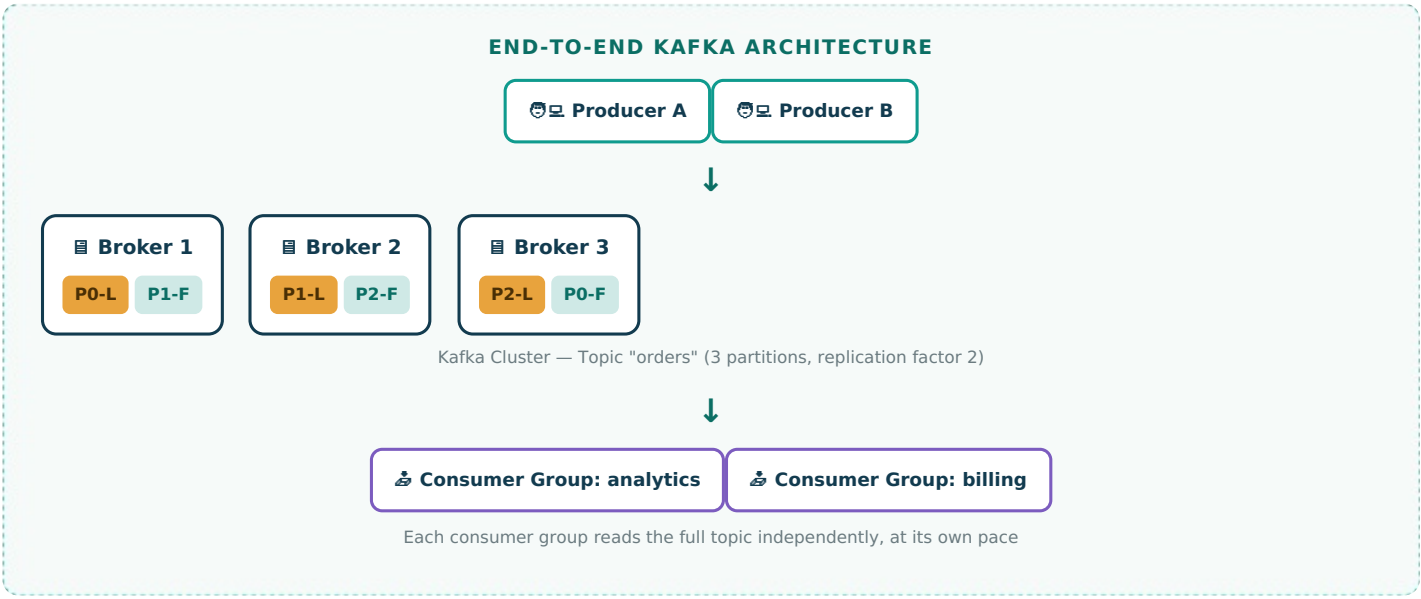
A: The leader handles all reads/writes; followers replicate the leader and can take over if it fails.

CHAPTER 3

🏗 Kafka Architecture

How all the pieces — producers, brokers, topics, partitions, and consumers — fit together.

Complete Architecture Diagram



Data Flow: Producer → Broker → Topic → Partition → Consumer

1. **Producer creates a message** and optionally assigns it a key (e.g., customer ID).
2. **Producer sends the message** to the Kafka cluster, targeting a topic (e.g., `orders`).
3. **Kafka decides the partition** — if a key was given, the same key always lands on the same partition (via hashing). If no key, Kafka spreads messages round-robin (or uses a "sticky" batching strategy in modern versions).
4. **The partition's leader broker** appends the message to its log and replicates it to follower brokers.
5. **Broker acknowledges** the write back to the producer (based on the `acks` setting).
6. **Consumers poll** the broker for new messages starting from their last committed offset.
7. **Consumer processes the message** and commits the new offset, marking it as read.

> **NOTE**
> Kafka is "pull-based" — consumers actively **poll** brokers for new data rather than brokers pushing data to consumers. This gives consumers full control over their own pace.

ZooKeeper (Legacy) vs. KRaft Mode (Modern Kafka)

Every distributed system needs a way to track cluster metadata (which brokers exist, who is the leader of each partition, cluster configuration). Historically, Kafka relied on an external coordination service called **Apache ZooKeeper** for this. Since **Kafka 3.3+ (KRaft mode is now the default and recommended for all new deployments as of Kafka 4.0)**, Kafka manages this metadata internally using its own Raft-based consensus protocol — no external ZooKeeper needed.

🗂 ZooKeeper Mode (Legacy)

- Requires a separate ZooKeeper cluster
- More moving parts to operate & monitor
- Slower controller failover
- Deprecated — removed in Kafka 4.0

⚡ KRaft Mode (Modern)

- Kafka brokers manage metadata themselves
- Simpler to deploy — one system, not two
- Faster failover, supports more partitions
- Default for all new Kafka installs today

✅ **BEST PRACTICE**

Always use **KRaft mode** for any new Kafka project or learning setup. ZooKeeper is legacy and being phased out — you will still see it referenced in older tutorials and existing production clusters, so it's worth understanding conceptually, but don't build new things on it.

🗒 Chapter 3 Summary

- Producers write to topics; Kafka routes messages into partitions (by key hash or round-robin).
- Brokers store partitions; each partition has one leader and zero or more followers.
- Consumers pull data at their own pace and track progress using offsets.
- Cluster metadata is managed by ZooKeeper (legacy) or KRaft (modern, recommended).

🧠 Mini Quiz

Q1. How does Kafka decide which partition a keyed message goes to?

A: It hashes the key — the same key always maps to the same partition.

Q2. Is Kafka push-based or pull-based for consumers?

A: Pull-based — consumers poll for new messages.

Q3. What replaced ZooKeeper in modern Kafka?

A: KRaft mode, Kafka's built-in Raft-based metadata management.

CHAPTER 4

⚙ How Kafka Works

The mechanics behind sending, reading, ordering, and retaining data.

Sending Messages

A producer batches messages in memory, serializes them (e.g., to JSON or Avro bytes), picks a target partition, and sends the batch to the partition's leader broker. The broker appends the batch to the end of the partition log — an append-only file on disk.

Reading Messages

Consumers issue a `poll()` request specifying the topic/partition and the offset to start from. The broker returns a batch of messages starting at that offset. The consumer processes them and then commits the new offset (either automatically or manually) so it knows where to resume next time.

Message Ordering

> 💡 **REMEMBER**
>
> Order is guaranteed **within a partition only**. If you need strict ordering for a specific entity (e.g., all events for one user), always send those events with the **same key** so they land in the same partition.

Partition Assignment

When a consumer group has multiple consumers, Kafka's group coordinator assigns each partition to exactly one consumer in the group using a **partition assignment strategy** (Range, RoundRobin, Sticky, or Cooperative Sticky). If a consumer joins or leaves, a **rebalance** occurs and partitions are reassigned.

Strategy	Behavior
Range	Assigns contiguous partition ranges per topic to each consumer
RoundRobin	Spreads all partitions evenly across all consumers
Sticky	Minimizes partition movement during rebalances
Cooperative Sticky	Modern default — rebalances incrementally, reducing downtime

Consumer Offsets

Kafka stores committed consumer offsets in an internal topic called `__consumer_offsets`. This means offset tracking is itself durable and fault-tolerant — if a consumer crashes and restarts, it resumes exactly where it left off (based on its last commit).

> ⚠ **IMPORTANT**
>
> **auto.offset.reset** determines what a consumer does when it has no committed offset (e.g., first run): `earliest` reads from the very start of the partition; `latest` only reads new messages from now on.

Data Retention

Kafka doesn't delete a message immediately after it's consumed (unlike traditional queues). Instead, every topic has a **retention policy** — messages are kept for a configured time (default 7 days) or until the topic reaches a size limit, whichever comes first. This lets multiple consumers replay the same data independently.

Setting	Purpose	Example
`retention.ms`	How long to keep messages	604800000 (7 days)
`retention.bytes`	Max size per partition before old data is dropped	1073741824 (1 GB)
`cleanup.policy=compact`	Keep only the latest value per key forever (for state topics)	used by Kafka Streams state stores

Fault Tolerance

Kafka survives broker failures through **replication**. If a leader broker crashes, one of its in-sync followers (an "ISR" — In-Sync Replica) is automatically elected as the new leader with no data loss, as long as `acks=all` was used when producing.

INTERVIEW TIP

A favorite interview question: "*What happens if the leader broker for a partition goes down?*" Good answer: "One of the in-sync replicas (ISR) is elected as the new leader automatically via the controller; as long as `acks=all` and `min.insync.replicas` are configured properly, no committed data is lost."

Chapter 4 Summary

- Producers append to partitions; consumers poll and commit offsets to track progress.
- Ordering only holds within a partition — use consistent keys for per-entity ordering.
- Consumer group coordinators assign partitions and rebalance when membership changes.
- Retention keeps data for a set time/size regardless of consumption, enabling replay.
- Replication + leader election gives Kafka automatic fault tolerance.

Mini Quiz

Q1. Where does Kafka store committed consumer offsets?

A: In the internal topic `__consumer_offsets`.

Q2. What does `auto.offset.reset=earliest` do?

A: Tells a consumer with no prior offset to start reading from the beginning of the partition.

Q3. Does Kafka delete a message as soon as it's consumed?

A: No — messages persist until the retention time/size limit is reached, regardless of consumption.

CHAPTER 5

🖫 Kafka Installation

Getting a real Kafka cluster running on your machine — three different ways.

> **📝 NOTE**
> All examples below use modern Kafka (KRaft mode) — no ZooKeeper required. Requires Java 11+ installed.

Install on Windows

1. Install Java (JDK 11+) and verify with `java -version`.
2. Download the latest Kafka binary (.tgz) from **kafka.apache.org/downloads**.
3. Extract it, e.g., to `C:\kafka`.
4. Open Command Prompt in the Kafka folder and generate a cluster ID + format storage:

CMD

```
:: Generate a unique cluster ID
.\bin\windows\kafka-storage.bat random-uuid

:: Format the storage directory using that ID
.\bin\windows\kafka-storage.bat format -t <uuid> -c .\config\kraft\server.properties

:: Start the Kafka broker
.\bin\windows\kafka-server-start.bat .\config\kraft\server.properties
```

Install on Ubuntu / Linux

BASH

```
# 1. Install Java
sudo apt update
sudo apt install openjdk-17-jdk -y

# 2. Download and extract Kafka
wget https://downloads.apache.org/kafka/3.8.0/kafka_2.13-3.8.0.tgz
tar -xzf kafka_2.13-3.8.0.tgz
cd kafka_2.13-3.8.0

# 3. Generate cluster ID and format storage (KRaft mode)
KAFKA_CLUSTER_ID="$(bin/kafka-storage.sh random-uuid)"
bin/kafka-storage.sh format -t $KAFKA_CLUSTER_ID -c config/kraft/server.properties

# 4. Start the broker
bin/kafka-server-start.sh config/kraft/server.properties
```

> **✅ BEST PRACTICE**
> Always check the official **kafka.apache.org/downloads** page for the current stable version number before downloading — version numbers in tutorials go stale fast.

Install Using Docker (Recommended for Beginners)

Docker is the fastest, cleanest way to get Kafka running without installing Java or managing config files manually.

DOCKER-COMPOSE.YML

```
version: '3.8'
services:
  kafka:
    image: apache/kafka:3.8.0
    container_name: kafka
    ports:
      - "9092:9092"
    environment:
      KAFKA_NODE_ID: 1
      KAFKA_PROCESS_ROLES: 'broker,controller'
      KAFKA_LISTENERS: 'PLAINTEXT://:9092,CONTROLLER://:9093'
      KAFKA_ADVERTISED_LISTENERS: 'PLAINTEXT://localhost:9092'
      KAFKA_CONTROLLER_QUORUM_VOTERS: '1@localhost:9093'
      KAFKA_CONTROLLER_LISTENER_NAMES: 'CONTROLLER'
      KAFKA_LISTENER_SECURITY_PROTOCOL_MAP: 'CONTROLLER:PLAINTEXT,PLAINTEXT:PLAINTEXT'
```

BASH

```
# Start the container
docker compose up -d

# Verify it's running
docker ps
```

Start Kafka in KRaft Mode (Manual Setup Recap)

Step	Command
1. Generate cluster ID	`kafka-storage.sh random-uuid`
2. Format storage	`kafka-storage.sh format -t <id> -c config/kraft/server.properties`
3. Start broker	`kafka-server-start.sh config/kraft/server.properties`

Start a Producer and Consumer (Quick Test)

BASH — TERMINAL 1

```
bin/kafka-topics.sh --create --topic test-topic --bootstrap-server localhost:9092
bin/kafka-console-producer.sh --topic test-topic --bootstrap-server localhost:9092
# Type messages and press Enter to send each one
```

BASH — TERMINAL 2

```
bin/kafka-console-consumer.sh --topic test-topic --from-beginning --bootstrap-server localhost:9092
# You'll see every message typed in Terminal 1 appear here
```

⚠ COMMON MISTAKE

Forgetting to open a firewall port or using `localhost` in `advertised.listeners` when connecting from another machine or a Docker container on a different network. Set `advertised.listeners` to an address reachable by your clients.

📋 Chapter 5 Summary

- Kafka needs Java installed; modern Kafka runs in KRaft mode (no ZooKeeper).
- Docker is the fastest path for beginners — one `docker compose up` and you're running.
- Native installs on Windows/Linux require generating a cluster ID and formatting storage first.
- Console producer/consumer scripts are the quickest way to sanity-check a running cluster.

CHAPTER 6

⌨ Kafka Command Line

The essential CLI tools every Kafka engineer uses daily.

Create a Topic

BASH

```
bin/kafka-topics.sh --create \
  --topic orders \
  --bootstrap-server localhost:9092 \
  --partitions 3 \
  --replication-factor 1
```

List Topics

BASH

```
bin/kafka-topics.sh --list --bootstrap-server localhost:9092
```

Describe a Topic

BASH

```
bin/kafka-topics.sh --describe --topic orders --bootstrap-server localhost:9092
```

Output shows partition count, replication factor, and which broker is leader/ISR for each partition.

Delete a Topic

BASH

```
bin/kafka-topics.sh --delete --topic orders --bootstrap-server localhost:9092
```

> ⚠ **COMMON MISTAKE**
>
> Deleting a topic is **irreversible** and instantly wipes all its data. Double-check the topic name, especially in shared/production clusters.

Produce Messages

BASH

```
# Simple producer
bin/kafka-console-producer.sh --topic orders --bootstrap-server localhost:9092

# Producer with a message key (key:value separated by a colon)
bin/kafka-console-producer.sh --topic orders --bootstrap-server localhost:9092 \
  --property "parse.key=true" --property "key.separator=:"
```

Consume Messages

BASH

```
# Read only new messages
bin/kafka-console-consumer.sh --topic orders --bootstrap-server localhost:9092

# Read from the very beginning
bin/kafka-console-consumer.sh --topic orders --from-beginning --bootstrap-server localhost:9092

# Show keys along with values
bin/kafka-console-consumer.sh --topic orders --bootstrap-server localhost:9092 \
  --property print.key=true --property key.separator=":"
```

Useful Administration Commands

Task	Command
List consumer groups	`kafka-consumer-groups.sh --list --bootstrap-server localhost:9092`
Describe a group (see lag)	`kafka-consumer-groups.sh --describe --group my-group --bootstrap-server localhost:9092`
Reset offsets to earliest	`kafka-consumer-groups.sh --group my-group --topic orders --reset-offsets --to-earliest --execute --bootstrap-server localhost:9092`
Change partition count	`kafka-topics.sh --alter --topic orders --partitions 6 --bootstrap-server localhost:9092`
Check broker configs	`kafka-configs.sh --describe --entity-type brokers --entity-name 1 --bootstrap-server localhost:9092`
View topic configs	`kafka-configs.sh --describe --entity-type topics --entity-name orders --bootstrap-server localhost:9092`

> **REMEMBER**
>
> You can **increase** a topic's partition count later, but you can **never decrease** it — plan partition counts with future growth in mind.

Chapter 6 Summary

- `kafka-topics.sh` creates, lists, describes, alters, and deletes topics.
- `kafka-console-producer.sh` / `kafka-console-consumer.sh` are quick manual testing tools.
- `kafka-consumer-groups.sh` is essential for monitoring consumer lag and resetting offsets.
- Partition counts can only go up, never down — plan ahead.

CHAPTER 7

🐍 Kafka with Python

Writing real producer and consumer applications using kafka-python.

Install kafka-python

BASH

```
pip install kafka-python
```

> **📝 NOTE**
>
> `confluent-kafka` (built on the C library `librdkafka`) is faster and more production-grade, while `kafka-python` is pure Python and easier for beginners to read and debug. Both use very similar APIs.

Producer Example

PYTHON

```
from kafka import KafkaProducer
import json

# Create a producer that connects to our local broker
producer = KafkaProducer(
    bootstrap_servers='localhost:9092',
    value_serializer=lambda v: json.dumps(v).encode('utf-8'),  # auto-convert dict -> JSON bytes
    key_serializer=lambda k: k.encode('utf-8') if k else None
)

# Build a sample order event
order = {"order_id": 1042, "customer": "Yousseg", "amount": 199.50}

# Send it to the "orders" topic, keyed by customer name
producer.send("orders", key="Yousseg", value=order)

producer.flush()  # make sure the message is actually sent before exiting
print("Order sent!")
```

Consumer Example

PYTHON

```
from kafka import KafkaConsumer
import json

# Create a consumer subscribed to the "orders" topic
consumer = KafkaConsumer(
    "orders",
    bootstrap_servers='localhost:9092',
    auto_offset_reset='earliest',      # start from the beginning if no offset saved
    enable_auto_commit=True,             # automatically commit offsets
    group_id="order-processors",        # consumer group name
    value_deserializer=lambda v: json.loads(v.decode('utf-8'))
)

# Loop forever, printing each new message as it arrives
for message in consumer:
    print(f"Received order: {message.value} (partition={message.partition}, offset={message.offset})")
```

Sending JSON Messages (Recap Pattern)

REMEMBER

Kafka only sends and stores **bytes**. Any structured data (dict, object) must be **serialized** before sending (e.g., to JSON) and **deserialized** after receiving. Always keep the producer's serializer and the consumer's deserializer in sync.

Error Handling

PYTHON

```python
from kafka import KafkaProducer
from kafka.errors import KafkaError
import json

producer = KafkaProducer(
    bootstrap_servers='localhost:9092',
    value_serializer=lambda v: json.dumps(v).encode('utf-8'),
    acks='all',        # wait for all in-sync replicas to confirm the write
    retries=5          # retry transient failures automatically
)

def on_send_success(record_metadata):
    print(f"Delivered to {record_metadata.topic} partition {record_metadata.partition}")

def on_send_error(excp):
    print(f"Failed to deliver message: {excp}")

try:
    future = producer.send("orders", value={"order_id": 1043})
    future.add_callback(on_send_success)
    future.add_errback(on_send_error)
except KafkaError as e:
    print(f"Kafka error: {e}")
```

Best Practices

Practice	Why
Set `acks='all'` for important data	Guarantees the message is replicated before being acknowledged
Always call `producer.flush()` before shutdown	Ensures buffered messages aren't lost
Use meaningful message keys	Preserves per-entity ordering and enables log compaction
Set `enable_auto_commit=False` for critical pipelines	Lets you commit offsets only after successful processing (avoids data loss on crash)
Catch and log `KafkaError` exceptions	Prevents silent message loss
Use a schema (Avro/JSON Schema) for production topics	Prevents producers and consumers from drifting out of sync on data format

BEST PRACTICE

In production pipelines, commit offsets **manually, after** your processing logic succeeds — not automatically on a timer. This avoids "losing" a message if your app crashes mid-processing after the offset was already auto-committed.

Chapter 7 Summary

- `kafka-python` gives you `KafkaProducer` and `KafkaConsumer` classes with simple, Pythonic APIs.
- Always serialize/deserialize data explicitly — Kafka only moves bytes.
- Use `acks='all'`, retries, and manual offset commits for reliability in real pipelines.

CHAPTER 8

🔥 Kafka with Apache Spark

Connecting Kafka streams to Spark Structured Streaming for real-time ETL.

Since you're already building PySpark skills, this is a natural next step: Spark can read directly from Kafka topics as a streaming DataFrame, apply transformations, and write results back out — to Kafka, Parquet, a database, or your medallion architecture layers.

Reading from Kafka

PYTHON — PySpark

```python
from pyspark.sql import SparkSession

# Create a SparkSession with the Kafka connector package included
spark = SparkSession.builder \
    .appName("KafkaSparkIntegration") \
    .config("spark.jars.packages", "org.apache.spark:spark-sql-kafka-0-10_2.12:3.5.0") \
    .getOrCreate()

# Read a stream of raw events from the "orders" topic
raw_df = spark.readStream \
    .format("kafka") \
    .option("kafka.bootstrap.servers", "localhost:9092") \
    .option("subscribe", "orders") \
    .option("startingOffsets", "earliest") \
    .load()

# Kafka messages arrive as raw bytes in "key" and "value" columns — cast to string first
events_df = raw_df.selectExpr("CAST(key AS STRING)", "CAST(value AS STRING)", "timestamp")
```

Structured Streaming Basics

PYTHON — PySpark

```python
from pyspark.sql.functions import from_json, col
from pyspark.sql.types import StructType, StringType, DoubleType, IntegerType

# Define the schema of the JSON payload inside the Kafka "value" field
order_schema = StructType() \
    .add("order_id", IntegerType()) \
    .add("customer", StringType()) \
    .add("amount", DoubleType())

# Parse the JSON string into structured columns
parsed_df = events_df.select(
    from_json(col("value"), order_schema).alias("data")
).select("data.*")
```

> **NOTE**
>
> Structured Streaming treats a Kafka topic as an **unbounded, ever-growing table**. Every micro-batch trigger, Spark processes only the new rows (messages) that have arrived since the last batch — you write your transformation logic exactly like you would for a static DataFrame.

Writing to Kafka

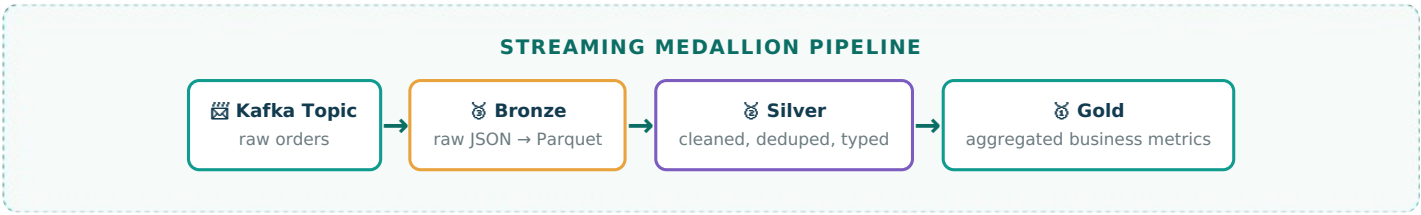
PYTHON — PySpark

```
# Prepare output — Kafka sink requires a "key" and "value" column
output_df = parsed_df.selectExpr(
    "CAST(customer AS STRING) AS key",
    "to_json(struct(*)) AS value"
)

query = output_df.writeStream \
    .format("kafka") \
    .option("kafka.bootstrap.servers", "localhost:9092") \
    .option("topic", "orders_enriched") \
    .option("checkpointLocation", "/tmp/checkpoints/orders_enriched") \
    .start()

query.awaitTermination()
```

Example ETL Pipeline: Bronze → Silver → Gold from Kafka



PYTHON — Bronze layer write example

```
bronze_query = parsed_df.writeStream \
    .format("parquet") \
    .option("path", "/data/bronze/orders") \
    .option("checkpointLocation", "/data/checkpoints/bronze_orders") \
    .outputMode("append") \
    .trigger(processingTime="30 seconds") \
    .start()
```

✓ BEST PRACTICE

Always set a **checkpointLocation** for streaming queries — it stores offset progress so Spark can resume exactly where it left off after a restart, without reprocessing or skipping data.

Chapter 8 Summary

- Spark reads Kafka topics as streaming DataFrames via `readStream.format("kafka")`.
- Kafka's raw `value` column is bytes — parse it with `from_json` and a defined schema.
- Writing back to Kafka requires "key" and "value" columns in the output DataFrame.
- Checkpointing makes streaming ETL pipelines fault-tolerant and resumable — essential for medallion-style bronze/silver/gold pipelines fed by Kafka.

CHAPTER 9

Kafka Ecosystem

The tools built around Kafka that make it a full data platform, not just a message log.

Kafka Connect

A framework for moving data **in and out** of Kafka without writing custom code. **Source connectors** pull data from external systems (databases, files, APIs) into Kafka topics. **Sink connectors** push data from Kafka topics into external systems (data warehouses, Elasticsearch, S3).

Kafka Streams

A Java/Scala **library** (not a separate cluster) for building stream-processing applications directly on top of Kafka — filtering, joining, aggregating, and transforming data in real time, reading from one topic and writing results to another.

Schema Registry

A central service that stores and enforces **schemas** (usually Avro, Protobuf, or JSON Schema) for the data flowing through Kafka topics. It prevents producers from sending malformed or incompatible data that would break downstream consumers.

> ⚠ **IMPORTANT**
>
> Without a Schema Registry, a small typo or an unannounced field change in a producer can silently break every downstream consumer. In production systems handling structured data, a schema registry is considered essential, not optional.

Confluent Platform

Confluent (founded by Kafka's original creators) offers a commercial platform built around open-source Kafka, adding: a managed cloud service, Schema Registry, ksqlDB (SQL-based stream processing), pre-built connectors, GUI monitoring (Control Center), and enterprise security features.

Tool	Category	What It's For
Kafka Connect	Integration	Move data in/out of Kafka without custom code
Kafka Streams	Processing	Build stream-processing apps as a Java/Scala library
ksqlDB	Processing	Stream processing using SQL syntax
Schema Registry	Governance	Enforce and version data schemas
Confluent Platform / Cloud	Commercial distribution	Managed Kafka + extra enterprise tooling

> 💡 **REMEMBER**
>
> Apache Kafka itself is free and open source. Confluent, Amazon MSK, and other vendors sell **managed hosting and extra tooling** around it — you are never required to use them to run Kafka.

Chapter 9 Summary

- Kafka Connect handles moving data in and out of Kafka via reusable connectors.
- Kafka Streams and ksqlDB let you process streaming data without a separate processing cluster.
- Schema Registry protects data quality across producers and consumers.
- Confluent Platform packages these tools plus enterprise features on top of open-source Kafka.

CHAPTER 10

⚡ Performance — Why Kafka Is Fast

The engineering decisions that let Kafka handle millions of messages per second.

Batching

Instead of sending one message at a time (expensive network round-trips), producers group messages into **batches** before sending. Fewer, larger network requests dramatically increase throughput.

Compression

Producers can compress batches (using `gzip`, `snappy`, `lz4`, or `zstd`) before sending. Smaller payloads mean less network bandwidth and less disk usage — brokers store and forward the compressed batch without decompressing it.

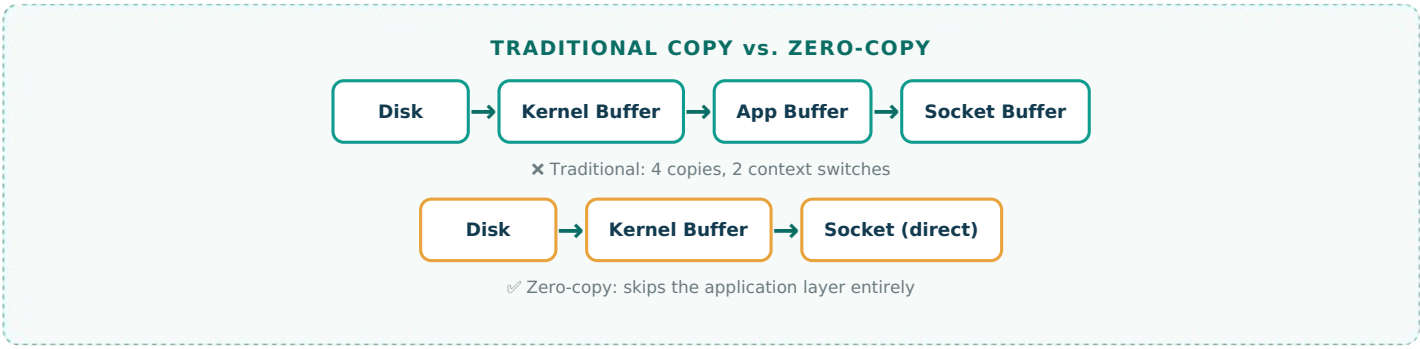
Sequential Disk Writes

Kafka partitions are append-only logs — new data is always written to the **end** of the file. Sequential writes are dramatically faster than random writes on both spinning disks and SSDs, because the disk head (or controller) doesn't need to jump around.

> **NOTE**
> This is counter-intuitive to many beginners: Kafka relies heavily on disk, not memory, for its speed — because sequential disk I/O can actually outperform random memory access patterns in other systems.

Zero-Copy

When sending data to consumers, Kafka uses the operating system's **zero-copy** mechanism (`sendfile()`) to transfer data straight from disk cache to the network socket — completely bypassing the application layer and avoiding unnecessary memory copies.



Replication

While replication adds some write overhead, it's designed to be efficient: followers fetch from the leader in the same batched, sequential-write, zero-copy manner as any consumer — replication doesn't require a fundamentally different (slower) code path.

Partitions and Scalability

Partitions are Kafka's core scaling unit. More partitions = more parallelism for both writing (more brokers involved) and reading (more consumers can work simultaneously). Throughput scales roughly linearly as you add partitions and brokers, up to a point.

Technique	Effect on Performance
Batching	Fewer network round-trips → higher throughput
Compression	Less network & disk I/O
Sequential disk writes	Near-memory-speed disk performance
Zero-copy transfer	Less CPU and memory overhead when serving consumers
Partitioning	Horizontal scalability across brokers and consumers

INTERVIEW TIP

If asked "Why is Kafka so fast?" — hit these five points in order: **sequential I/O, batching, compression, zero-copy, and partition-based parallelism.** This is one of the most common Kafka interview questions.

Chapter 10 Summary

- Kafka's speed comes from architectural choices, not raw hardware: sequential writes, batching, compression, zero-copy, and partitioned parallelism.
- These design choices let a single Kafka cluster handle millions of messages per second on commodity hardware.

CHAPTER 11

✔ Best Practices

Practical guidelines for designing real Kafka systems that scale and stay reliable.

Topic Naming Conventions

Convention	Example
`<domain>.<entity>.<event>`	`ecommerce.orders.created`
Lowercase, dot or hyphen separated	`user-signups`, not `UserSignUps`
Include environment for non-prod clusters	`dev.orders.created`

Choosing the Number of Partitions

Base it on your **target throughput** and the **number of consumers** you expect to run in parallel. A common starting formula: partitions ≥ max expected consumers in a single consumer group.

> ✔ **BEST PRACTICE**
>
> Start with a moderate number (e.g., 6–12 for most learning/small projects) — you can always increase later, but never decrease. Too many partitions increases overhead (open file handles, replication traffic, leader election time).

Replication Factor

Environment	Recommended Replication Factor
Local learning / single broker	1
Small production cluster	3
Critical financial/data systems	3+ with `min.insync.replicas=2`

Message Key Selection

- Use a key when you need **ordering per entity** (e.g., all events for one customer in order).
- Avoid keys with extremely uneven distribution (e.g., a boolean flag) — this creates "hot" partitions.
- No key at all is fine when strict ordering isn't required — Kafka spreads load evenly.

Consumer Group Design

- One consumer group per independent "use case" of the data (e.g., `billing-service`, `analytics-service`).
- Keep consumers within a group stateless where possible to simplify scaling and rebalancing.
- Match consumer count to partition count for full parallelism, no idle consumers.

Monitoring Kafka

Metric	Why It Matters
Consumer lag	How far behind a consumer group is — the #1 health signal
Under-replicated partitions	Signals broker or network problems
Request latency (produce/fetch)	Detects performance degradation
Disk usage per broker	Prevents brokers from running out of space
ISR shrink/expand rate	Indicates replication instability

Common tools: Prometheus + Grafana with the Kafka Exporter, Confluent Control Center, Burrow (for lag monitoring).

⚠ **COMMON MISTAKE**

Treating "consumer is running" as "consumer is healthy." A consumer can be alive but falling further behind every second (growing lag) — always monitor lag directly, not just process uptime.

Chapter 11 Summary

- Use consistent, structured topic naming from day one.
- Size partitions for your expected parallelism; replication factor 3 is the production standard.
- Choose message keys deliberately based on ordering needs, avoiding hot partitions.
- Design consumer groups around use cases, and always monitor consumer lag.

CHAPTER 12

💼 Common Interview Questions

Beginner, intermediate, and scenario-based questions with full answers.

Beginner-Level Questions

Q1. What is Apache Kafka, in one sentence?

A distributed event streaming platform that lets applications publish, store, and subscribe to real-time streams of data reliably and at scale.

Q2. What is the difference between a topic and a partition?

A topic is a logical category of messages; a partition is a physically ordered sub-log that a topic is split into, enabling parallelism and scale.

Q3. What is an offset?

A unique, sequential ID assigned to each message within a partition, used by consumers to track their read position.

Q4. What's the difference between a Kafka queue and a topic?

Kafka doesn't remove messages after they're read like a traditional queue. Multiple consumer groups can independently read the full topic, and data persists based on retention policy, not consumption.

Q5. What is a consumer group?

A set of consumers that cooperate to read a topic, with each partition assigned to only one consumer within the group at a time.

Intermediate-Level Questions

Q6. What does the "acks" producer setting control?

How many broker replicas must acknowledge a write before the producer considers it successful: `acks=0` (no wait, fastest, risk of loss), `acks=1` (leader only), `acks=all` (all in-sync replicas, safest).

Q7. What is a rebalance, and what triggers one?

A reassignment of partitions among consumers in a group. Triggered when a consumer joins, leaves, crashes, or a topic's partition count changes.

Q8. What is an In-Sync Replica (ISR)?

A follower replica that is fully caught up with the leader's log and eligible to be promoted to leader if the current leader fails.

Q9. What is the difference between at-most-once, at-least-once, and exactly-once delivery?

At-most-once: message may be lost, never duplicated. At-least-once: message never lost, may be duplicated. Exactly-once: message processed exactly one time (Kafka supports this via idempotent producers + transactions).

Q10. What is log compaction?

A retention strategy that keeps only the latest value for each message key forever, deleting older values for the same key — used for topics representing current "state" rather than an event history.

Scenario-Based Questions

Q11. Your consumer group's lag keeps growing even though the consumers are running. What could be wrong, and how do you investigate?

Possible causes: consumers are too slow to process each message (heavy processing logic), too few consumers relative to partitions, a downstream dependency (DB, API) is slow, or a "poison pill" message causing repeated retries. Investigate with `kafka-consumer-groups.sh --describe` to see per-partition lag, check consumer application logs and processing time per message, and confirm partition-to-consumer assignment is balanced.

Q12. You need strict ordering of events per customer, but also need to scale to many consumers. How do you design this?

Use the customer ID as the message key so all events for one customer always land on the same partition (preserving order), and create enough partitions to allow one consumer per partition for parallelism across different customers.

Q13. A topic partition's leader broker just crashed. Walk through what happens next.

The cluster controller detects the failure, selects an in-sync replica as the new leader, updates metadata so producers/consumers discover the new leader automatically, and (assuming `acks=all` was used) no committed messages are lost.

Q14. How would you design a Kafka topic strategy for an e-commerce order pipeline feeding both real-time fraud detection and a nightly batch warehouse load?

Use one raw `orders.created` topic as the single source of truth, consumed independently by two separate consumer groups: a low-latency fraud-detection service, and a batch-loading job (e.g., Spark) that reads periodically or continuously and lands data into the warehouse — each group tracks its own offsets without interfering with the other.

REMEMBER

Interviewers care less about memorized definitions and more about whether you can reason through **trade-offs** (throughput vs. durability, ordering vs. parallelism). Practice explaining the "why," not just the "what."

Chapter 12 Summary

- Beginner questions test vocabulary; intermediate questions test mechanics (acks, ISR, rebalances, compaction).
- Scenario questions test design thinking — always talk through trade-offs out loud.
- Delivery guarantees (at-most-once, at-least-once, exactly-once) are asked in nearly every Kafka interview.

CHAPTER 13

🛠 Hands-on Projects

Six portfolio-ready projects, ordered from simplest to most advanced.

#	Project	Skills Practiced	Difficulty
1	Log Processing System	Producers, topics, console consumers	🟢 Beginner
2	Real-Time Order Processing	Python producer/consumer, keys, consumer groups	🟢 Beginner
3	Clickstream Analytics	Multiple topics, aggregation windows	🟡 Intermediate
4	IoT Sensor Pipeline	High-throughput producers, partitioning strategy	🟡 Intermediate
5	Spark + Kafka Streaming ETL	Structured Streaming, medallion architecture	🔴 Advanced
6	Airflow + Kafka Orchestration	Batch/stream hybrid pipelines, scheduling	🔴 Advanced

1. Log Processing System 🟢

Goal: Simulate an application generating logs, stream them through Kafka, and consume/filter them by severity.

- Create a topic `app-logs` with 3 partitions.
- Write a Python script that generates fake log lines (INFO/WARN/ERROR) and sends them as producer messages.
- Write a consumer that filters and prints only ERROR-level logs in real time.
- **Stretch goal:** Write ERROR logs to a separate topic using a second producer inside the consumer (a simple "router" pattern).

2. Real-Time Order Processing 🟢

Goal: Build the classic e-commerce pipeline — orders created, validated, and confirmed.

- Topic `orders.created` — producer sends random synthetic orders (order_id, customer, amount, items).
- Consumer 1 ("validator") checks for suspicious amounts and republishes to `orders.validated`.
- Consumer 2 ("notifier") reads `orders.validated` and simulates sending a confirmation message.
- Use the customer ID as the message key to preserve per-customer ordering.

3. Clickstream Analytics 🟡

Goal: Track user website clicks and compute rolling metrics.

- Topic `clickstream` — simulate page views with user_id, page, timestamp.
- Use PySpark Structured Streaming to compute "clicks per page per 1-minute window."
- Write aggregated results to console or a `clickstream_metrics` topic.

4. IoT Sensor Pipeline 🟡

Goal: Simulate hundreds of IoT devices streaming temperature data.

- Topic `sensor-readings` — key by `device_id` to keep each device's readings ordered.
- Producer simulates 100+ devices publishing readings every second (use threads or async).
- Consumer flags any reading above a threshold as an "alert" and writes it to `sensor-alerts`.

5. Spark + Kafka Streaming ETL 🔴

Goal: Build a full streaming medallion pipeline, matching your existing bronze/silver/gold work.

- Kafka topic → Bronze (raw Parquet, append mode).
- Bronze → Silver (deduplicate, cast types, handle nulls) using Structured Streaming.
- Silver → Gold (aggregate business metrics, e.g., revenue per hour) written to a queryable sink.
- Use checkpointing at every stage for fault tolerance.

6. Airflow + Kafka Orchestration

Goal: Combine streaming ingestion with scheduled batch processing.

- Kafka continuously ingests raw events into a bronze Parquet layer (via a long-running Spark Structured Streaming job).
- An Airflow DAG runs nightly to promote bronze → silver → gold using batch Spark jobs.
- Airflow sensors/tasks can also monitor consumer lag or trigger alerts if a stream stalls.

> ✅ **BEST PRACTICE — PORTFOLIO TIP**
>
> For each project, produce: a GitHub repo with a clean README, an architecture diagram, sample data, and a short write-up of design decisions (partition count, key choice, retention). This is exactly what data engineering interviewers want to see.

🛍 Chapter 13 Summary

- Start with simple producer/consumer scripts before adding Spark or Airflow.
- Each project reinforces a specific concept: keys/ordering, consumer groups, streaming ETL, orchestration.
- Document your projects like production systems — this is what makes a portfolio stand out.

CHAPTER 14

🐞 Common Errors & Troubleshooting

The errors every Kafka beginner hits — and exactly how to fix them.

"Topic Not Found" / UnknownTopicOrPartitionException

Cause	Fix
Topic wasn't created and auto-creation is disabled	Create it explicitly with `kafka-topics.sh --create`
Typo in topic name	Run `--list` to confirm the exact name
Connecting to the wrong cluster/broker	Double-check `bootstrap.servers`

"Broker Unavailable" / NoBrokersAvailable

Cause	Fix
Kafka isn't running	Check the broker process/container with `docker ps` or process list
Wrong host/port in `bootstrap.servers`	Verify the address matches your broker's actual listener config
Firewall or network blocking the port	Confirm port 9092 is open and reachable
Wrong `advertised.listeners` (e.g., "localhost" from inside Docker)	Set it to an address reachable by the client, not just the broker itself

Consumer Lag

Cause	Fix
Consumer processing logic too slow	Optimize processing, or scale out more consumers (up to partition count)
Too few partitions for the workload	Increase partitions (never decrease) to allow more parallel consumers
Downstream dependency (DB/API) is slow	Add batching, async calls, or a buffer/queue on the consumer side

Offset Problems (Data Reprocessed or Skipped)

Symptom	Likely Cause	Fix
Duplicate processing after restart	Offset committed before processing finished (auto-commit)	Commit offsets manually, after successful processing
Missing data after restart	Offset committed but processing crashed/failed silently	Add proper error handling; don't swallow exceptions before commit
Consumer starts from wrong point unexpectedly	Offset expired due to retention, or group ID changed	Check `offsets.retention.minutes` and keep consistent group IDs

Serialization Issues

Symptom	Cause	Fix
`SerializationException`	Producer serializer doesn't match the actual data type sent	Match serializer to data type (e.g., don't use StringSerializer for raw dict objects)
Consumer throws on `json.loads()`	Producer sent non-JSON or malformed JSON bytes	Validate data before sending; wrap deserialization in try/except and route bad messages to a dead-letter topic
Schema Registry compatibility error	Producer schema is incompatible with registered schema	Follow schema evolution rules (e.g., only add optional fields with defaults)

⚠ **COMMON MISTAKE**

Letting a single malformed message ("poison pill") crash your consumer in an infinite retry loop. Always wrap message processing in a try/except and route unparseable messages to a **dead-letter topic** instead of blocking the whole partition.

✅ **BEST PRACTICE — GENERAL TROUBLESHOOTING WORKFLOW**

1) Check broker health/logs first. 2) Confirm topic exists and has the expected partition count. 3) Check consumer group lag. 4) Check client-side logs for serialization/connection errors. 5) Reproduce with the console producer/consumer to isolate whether it's a broker issue or an application code issue.

📋 Chapter 14 Summary

- Most "topic not found" and "broker unavailable" errors are configuration/connectivity issues.
- Consumer lag is fixed by scaling consumers (up to partition count) or speeding up processing.
- Manual, post-processing offset commits prevent both data loss and duplicate processing.
- Always defend against malformed messages with try/except and dead-letter topics.

CHAPTER 15

⚖ Kafka vs. Other Messaging Systems

Knowing when Kafka is the right tool — and when it isn't.

Kafka vs. RabbitMQ

Aspect	Kafka	RabbitMQ
Model	Distributed log (pull-based)	Traditional message broker/queue (push-based)
Message retention	Retained after consumption (time/size based)	Deleted once acknowledged by default
Throughput	Very high (millions/sec)	High, but generally lower than Kafka
Ordering	Guaranteed per partition	Guaranteed per queue
Best for	Event streaming, log aggregation, replayable pipelines	Complex routing, task queues, RPC-style messaging

Kafka vs. ActiveMQ

Aspect	Kafka	ActiveMQ
Protocol support	Kafka protocol (custom)	JMS, AMQP, MQTT, STOMP (broad standard support)
Scalability	Built for horizontal scale from the ground up	Traditional broker, scales less naturally
Use case fit	High-volume streaming and event-driven architectures	Enterprise integration needing standard protocols

Kafka vs. Amazon Kinesis

Aspect	Kafka	Amazon Kinesis
Hosting	Self-managed or via any vendor (Confluent, MSK, etc.)	Fully managed AWS-only service
Ecosystem	Huge open-source ecosystem, cloud-agnostic	Deep native integration with other AWS services
Scaling	Manual partition management (or auto-scaling tools)	Shard-based, can auto-scale within AWS tooling
Cost model	Infrastructure cost (self-managed) or vendor pricing	Pay-per-shard-hour, simpler for AWS-centric teams

Kafka vs. Apache Pulsar

Aspect	Kafka	Apache Pulsar
Architecture	Storage and serving combined in the broker	Separates compute (brokers) from storage (BookKeeper)
Multi-tenancy	Possible but less native	Built-in multi-tenancy and geo-replication
Maturity / ecosystem	Much larger community, more tools and tutorials	Smaller but growing, technically elegant
Best for	Most general streaming use cases — the industry default	Multi-tenant SaaS platforms needing storage/compute separation

💡 **REMEMBER**

For interviews and most real-world jobs, **Kafka is still the default choice** for event streaming due to its ecosystem size, community, and hiring demand — knowing the alternatives well enough to explain trade-offs is usually more valuable than deep expertise in a niche competitor.

Chapter 15 Summary

- RabbitMQ/ActiveMQ excel at flexible routing and standard protocols; Kafka excels at high-throughput, replayable event streams.
- Kinesis is the AWS-managed equivalent, trading flexibility for operational simplicity.
- Pulsar offers a more modern storage/compute-separated architecture but with a smaller ecosystem.

CHAPTER 16

🗺 Learning Roadmap

Your path from "never touched Kafka" to portfolio-ready data engineer.

🟢 STAGE 1 — Beginner (Weeks 1–2)

Understand core concepts (topics, partitions, offsets, producers, consumers). Install Kafka via Docker. Run the console producer/consumer. Complete Project 1 (Log Processing) and Project 2 (Order Processing).

🟡 STAGE 2 — Intermediate (Weeks 3–5)

Write real producer/consumer apps in Python with proper serialization, keys, and error handling. Learn consumer groups and rebalancing deeply. Complete Project 3 (Clickstream) and Project 4 (IoT Pipeline). Start monitoring consumer lag.

🔴 STAGE 3 — Advanced (Weeks 6–9)

Integrate Kafka with PySpark Structured Streaming. Build a full streaming medallion pipeline (Project 5). Learn Kafka Connect and Schema Registry. Add Airflow orchestration (Project 6).

🏆 STAGE 4 — Interview-Ready (Weeks 10+)

Review Chapter 12's questions until you can explain trade-offs without notes. Polish your GitHub portfolio with README diagrams. Practice explaining your projects out loud, focusing on design decisions (partition count, key choice, delivery guarantees).

Recommended Projects, In Order

1. Log Processing System
2. Real-Time Order Processing (with keys + consumer groups)
3. Clickstream Analytics (PySpark windowed aggregation)
4. IoT Sensor Pipeline (high-throughput, partitioning strategy)
5. Spark + Kafka Streaming ETL (full medallion architecture)
6. Airflow + Kafka Orchestration (batch + streaming hybrid)

Recommended Tools to Learn After Kafka

Tool	Why It Pairs Well with Kafka
Apache Airflow	Orchestrates batch jobs around streaming pipelines
Apache Spark Structured Streaming	Processes Kafka streams at scale for ETL
dbt	Transforms data landed from Kafka into analytics-ready models
Docker & Kubernetes	Deploying and scaling Kafka-based services in production
Schema Registry (Avro/Protobuf)	Production-grade data contracts
Prometheus + Grafana	Monitoring Kafka clusters and consumer lag

✓ BEST PRACTICE

Don't try to learn everything at once. Master Chapters 1–7 solidly before touching Spark integration — a shaky grasp of partitions, offsets, and consumer groups will make every advanced topic harder than it needs to be.

★ FINAL CHEAT SHEET

📌 Apache Kafka — Everything on One Page

Pin this to your desk. Everything you learned, condensed.

Core Concepts

Term	Definition
Event / Message	A record of something that happened, with key, value, timestamp
Topic	Named category/stream of messages
Partition	Ordered sub-log of a topic; unit of parallelism
Offset	Sequential position of a message within a partition
Producer	Writes messages to a topic
Consumer	Reads messages from a topic
Consumer Group	Consumers sharing the work of reading a topic
Broker	A single Kafka server
Cluster	Group of brokers working together
Leader / Follower	Active partition copy vs. backup copies
ISR	In-Sync Replica — fully caught-up follower, eligible for leader election

Essential CLI Commands

BASH

```bash
# Create topic
kafka-topics.sh --create --topic X --partitions 3 --replication-factor 1 --bootstrap-server localhost:9092

# List / describe / delete
kafka-topics.sh --list --bootstrap-server localhost:9092
kafka-topics.sh --describe --topic X --bootstrap-server localhost:9092
kafka-topics.sh --delete --topic X --bootstrap-server localhost:9092

# Produce / consume
kafka-console-producer.sh --topic X --bootstrap-server localhost:9092
kafka-console-consumer.sh --topic X --from-beginning --bootstrap-server localhost:9092

# Consumer group lag
kafka-consumer-groups.sh --describe --group G --bootstrap-server localhost:9092
```

Python Quick Reference

PYTHON

```python
# Producer
producer = KafkaProducer(bootstrap_servers='localhost:9092',
    value_serializer=lambda v: json.dumps(v).encode())
producer.send("topic", key="k".encode(), value={"a":1})
producer.flush()

# Consumer
consumer = KafkaConsumer("topic", bootstrap_servers='localhost:9092',
    group_id="g", auto_offset_reset='earliest',
    value_deserializer=lambda v: json.loads(v.decode()))
for msg in consumer: print(msg.value)
```

Why Kafka Is Fast (5 Reasons)

Delivery Guarantees

Guarantee	Meaning
At-most-once	May lose messages, never duplicates
At-least-once	Never loses messages, may duplicate
Exactly-once	Never lost, never duplicated (idempotent producer + transactions)

Sizing Cheat Sheet

Setting	Rule of Thumb
Partitions	≥ max parallel consumers you'll ever run in one group
Replication Factor	1 (local/learning), 3 (production standard)
acks	'all' for critical data, '1' for high-throughput/low-criticality
Retention	7 days default; longer for replay/audit needs

Kafka vs. Alternatives — One-Line Verdicts

System	Choose It When...
RabbitMQ	You need complex routing or classic task queues
ActiveMQ	You need broad standard protocol support (JMS/AMQP/MQTT)
Amazon Kinesis	You're fully committed to AWS and want a managed service
Apache Pulsar	You need native multi-tenancy and storage/compute separation
Kafka	Default choice for high-throughput, replayable event streaming

FINAL REMINDER

Kafka mastery is built by **doing**, not reading. Spin up your Docker container, create a topic, and send your first message today.

You've completed the full guide!

- You now understand Kafka's core concepts, architecture, and internals.
- You can install Kafka, use the CLI, and build producer/consumer apps in Python.
- You know how Kafka integrates with Spark and the broader ecosystem.
- You're equipped with best practices, troubleshooting steps, and interview answers.
- Next step: pick Project 1 from Chapter 13 and start building.